

CO3-AT3-Assignment-2

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QUESTIONS

Q1

✓

Producer-Consumer 🔗 Inter-Thread Communication

25 marks

Q2

✓

Hospital Patient Registration 🔗 Multiple Exception Handling

25 marks

Q3

✓

Hospital Resource Allocation 🔗 Synchronization and Communication

25 marks

Q4

✓

Food Ordering System 🔗 Multiple Threads

25 marks

4/4 answered

Q1 Producer-Consumer 🔗 Inter-Thread Communication

25 marks

Problem Statement:

Develop a Java program to implement a Producer-Consumer scenario using inter-thread communication.

Requirements:

Create a producer thread and a consumer thread.

Use a shared buffer.

Use wait() and notify() for communication.

Ensure that the consumer waits when the buffer is empty.

Ensure that the producer waits when the buffer is full.

Your Code

```
1 public class ProducerConsumerDemo {
2     private int buffer;
3     private boolean available = false;
4     synchronized void produce(int value) {
5         try {
6             while (available)
7                 wait();
8             buffer = value;
9             available = true;
10            System.out.println("Produced: " + value);
11            notify();
12        } catch (InterruptedException e) {
13            // Handle exception
14        }
15    }
16 }
```

Input

stdin input

Output

```
Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
```




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25 marks

Q4

Food Ordering System ⓘ

Multiple Threads

25 marks

✓

4/4 answered

Your Code

```
1 public class ProducerConsumerDemo {
2     private int buffer;
3     private boolean available = false;
4     synchronized void produce(int value) {
5         try {
6             while (available)
7                 wait();
8             buffer = value;
9             available = true;
10            System.out.println("Produced: " + value);
11            notify();
12        } catch (InterruptedException e) {
13            Thread.currentThread().interrupt();
14        }
15    }
16    synchronized void consume() {
17        try {
18            while (!available)
19                wait();
20            System.out.println("Consumed: " + buffer);
21            available = false;
22            notify();
23        } catch (InterruptedException e) {
24            Thread.currentThread().interrupt();
25        }
26    }
27    public static void main(String[] args) {
28        ProducerConsumerDemo pc = new ProducerConsumerDemo();
29        Thread producer = new Thread(() -> {
30            for (int i = 1; i <= 5; i++)
31                pc.produce(i);
```

Input

stdin input

Output

```
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
```



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```
22         notify();
23     } catch (InterruptedException e) {
24         Thread.currentThread().interrupt();
25     }
26 }
27 public static void main(String[] args) {
28     ProducerConsumerDemo pc = new ProducerConsumerDemo();
29     Thread producer = new Thread(() -> {
30         for (int i = 1; i <= 5; i++)
31             pc.produce(i);
32     });
33     Thread consumer = new Thread(() -> {
34         for (int i = 1; i <= 5; i++)
35             pc.consume();
36     });
37     consumer.start();
38     producer.start();
39     try {
40         producer.join();
41         consumer.join();
42     } catch (InterruptedException e) {
43         Thread.currentThread().interrupt();
44     }
45 }
46 }
```

Run

Save

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QUESTIONS

Q1

Producer-Consumer ⚡ Inter-Thread Communication

25 marks

✓

Q2

Hospital Patient Registration ⚡ Multiple Exception Handling

25 marks

✓

Q3

Hospital Resource Allocation ⚡ Synchronization and Communication

25 marks

✓

Q4

Food Ordering System ⚡ Multiple Threads

25 marks

✓

4/4 answered

Q2 Hospital Patient Registration ⚡ Multiple Exception Handling

25 marks

Problem Statement:

Develop a Java program for hospital patient registration. The system should accept patient age number of visits and patient ID. Handle invalid input and invalid calculations using appropriate exception handling mechanisms.

Requirements:

Accept patient details.

Handle NumberFormatException and ArithmeticException.

Use multiple catch blocks.

Display meaningful error messages.

Use a finally block to indicate completion of registration.

Your Code

```
1 import java.util.Scanner;
2 public class HospitalPatientRegistration {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         try {
6             int id = Integer.parseInt(sc.nextLine());
7             int age = Integer.parseInt(sc.nextLine());
```

Input

101

45

5

Output

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Q4

Food Ordering System ⓘ

Multiple Threads

25 marks

4/4 answered

Your Code

```
1 import java.util.Scanner;
2 public class HospitalPatientRegistration {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         try {
6             int id = Integer.parseInt(sc.nextLine());
7             int age = Integer.parseInt(sc.nextLine());
8             int visits = Integer.parseInt(sc.nextLine());
9             int average = 100 / visits;
10            System.out.println("\nPatient ID: " + id);
11            System.out.println("Patient Age: " + age);
12            System.out.println("Number of Visits: " + visits);
13            System.out.println("Average: " + average);
14        } catch (NumberFormatException e) {
15            System.out.println("Error: Please enter valid numeric value");
16        } catch (ArithmeticException e) {
17            System.out.println("Error: Number of visits cannot be zero.");
18        } finally {
19            System.out.println("Patient registration completed.");
20        }
21        sc.close();
22    }
23 }
```

▶ Run

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Input

101
45
5

Output

Patient ID: 101
Patient Age: 45
Number of Visits: 5
Average: 20
Patient registration completed.

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QUESTIONS

Q1
Producer-Consumer ⚡ Inter-Thread Communication
25 marks

Q2
Hospital Patient Registration ⚡ Multiple Exception Handling
25 marks

Q3
Hospital Resource Allocation ⚡ Synchronization and Communication
25 marks

Q4
Food Ordering System ⚡ Multiple Threads
25 marks

4/4 answered

Q3 Hospital Resource Allocation ⚡ Synchronization and Communication25 marks

Problem Statement:
Develop a Java program to simulate allocation of a limited number of hospital resources such as ventilators beds or medical equipment to multiple patients.

Requirements:
Create multiple patient/request threads.
Maintain a shared number of available resources.
Use synchronization to control resource allocation.
Use inter-thread communication when resources become unavailable.
Release resources after use and notify waiting threads.
Handle invalid resource requests using appropriate exception handling.

Your Code

```
1 public class HospitalResourceAllocation {  
2     private int resources = 3;  
3     synchronized void requestResource(String patient) {  
4         try {  
5             while (resources <= 0) {
```

Input

stdin input

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Q4

Food Ordering System 🔒

Multiple Threads

25 marks

4/4 answered

Your Code

```
1 public class HospitalResourceAllocation {
2     private int resources = 3;
3     synchronized void requestResource(String patient) {
4         try {
5             while (resources <= 0) {
6                 System.out.println(patient + " is waiting for a resource");
7                 wait();
8             }
9             resources--;
10            System.out.println(patient + " allocated a resource. Available: " + resources);
11        } catch (InterruptedException e) {
12            Thread.currentThread().interrupt();
13            System.out.println(patient + " was interrupted.");
14        }
15    }
16    synchronized void releaseResource(String patient) {
17        resources++;
18        System.out.println(patient + " released a resource. Available: " + resources);
19        notifyAll();
20    }
21    public static void main(String[] args) {
22        HospitalResourceAllocation hospital = new HospitalResourceAllocation();
23        Thread p1 = new Thread(() -> {
24            hospital.requestResource("Patient 1");
25            try {
26                Thread.sleep(1000);
```

Input

stdin input

Output

```
Patient 1 allocated a resource.
Available: 2
Patient 4 allocated a resource.
Available: 1
Patient 3 allocated a resource.
Available: 0
```

```
27         } catch (InterruptedException e) {
28             Thread.currentThread().interrupt();
29         }
30         hospital.releaseResource("Patient 1");
31     });
32     Thread p2 = new Thread(() -> {
33         hospital.requestResource("Patient 2");
34         try {
35             Thread.sleep(1000);
36         } catch (InterruptedException e) {
37             Thread.currentThread().interrupt();
38         }
39         hospital.releaseResource("Patient 2");
40     });
41     Thread p3 = new Thread(() -> {
42         hospital.requestResource("Patient 3");
43         try {
44             Thread.sleep(1000);
45         } catch (InterruptedException e) {
46             Thread.currentThread().interrupt();
47         }
48         hospital.releaseResource("Patient 3");
49     });
50     Thread p4 = new Thread(() -> {
51         hospital.requestResource("Patient 4");
52         try {
53             Thread.sleep(1000);
54         } catch (InterruptedException e) {
```



```
45         } catch (InterruptedException e) {
46             Thread.currentThread().interrupt();
47         }
48         hospital.releaseResource("Patient 3");
49     });
50     Thread p4 = new Thread(() -> {
51         hospital.requestResource("Patient 4");
52         try {
53             Thread.sleep(1000);
54         } catch (InterruptedException e) {
55             Thread.currentThread().interrupt();
56         }
57         hospital.releaseResource("Patient 4");
58     });
59     p1.start();
60     p2.start();
61     p3.start();
62     p4.start();
63 }
64 }
```

 Run Save

Output

```
Patient 1 allocated a resource.
```

```
Available: 2
```

```
Patient 4 allocated a resource.
```

```
Available: 1
```

```
Patient 3 allocated a resource.
```

```
Available: 0
```


Output

```
Patient 2 is waiting for a  
resource...
```

```
Patient 1 released a resource.
```

```
Available: 1
```

```
Patient 2 allocated a resource.
```

```
Available: 0
```

Output

```
Patient 3 released a resource.
```

```
Available: 1
```

```
Patient 4 released a resource.
```

```
Available: 2
```

```
Patient 2 released a resource.
```

```
Available: 3
```


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Producer-Consumer ⚡ Inter-Thread Communication
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Q2
Hospital Patient Registration ⚡ Multiple Exception Handling
25 marks

Q3
Hospital Resource Allocation ⚡ Synchronization and Communication
25 marks

Q4
Food Ordering System ⚡ Multiple Threads
25 marks

4/4 answered

Q4 Food Ordering System ⚡ Multiple Threads

25 marks

Problem Statement:

Develop a multithreaded Java program for a food ordering system. Separate threads should process order placement food preparation and delivery.

Requirements:

Create three separate threads.

Each thread should perform its respective task.

Display the execution status of each task.

Demonstrate concurrent execution.

Your Code

```
1 public class FoodOrderingSystem {
2     public static void main(String[] args) {
3         Thread order = new Thread(() -> {
4             System.out.println("Order placement started");
5             try {
6                 Thread.sleep(500);
7             } catch (InterruptedException e) {
8                 Thread.currentThread().interrupt();

```

Input

stdin input

Output

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Communication

25 marks

Q4

Food Ordering System

Multiple Threads

25 marks

4/4 answered

Your Code

```
1 public class FoodOrderingSystem {
2     public static void main(String[] args) {
3         Thread order = new Thread(() -> {
4             System.out.println("Order placement started");
5             try {
6                 Thread.sleep(500);
7             } catch (InterruptedException e) {
8                 Thread.currentThread().interrupt();
9             }
10            System.out.println("Order placed successfully");
11        });
12        Thread preparation = new Thread(() -> {
13            System.out.println("Food preparation started");
14            try {
15                Thread.sleep(700);
16            } catch (InterruptedException e) {
17                Thread.currentThread().interrupt();
18            }
19            System.out.println("Food prepared successfully");
20        });
21        Thread delivery = new Thread(() -> {
22            System.out.println("Delivery started");
23            try {
24                Thread.sleep(600);
25            } catch (InterruptedException e) {
26                Thread.currentThread().interrupt();
```

Input

stdin input

Output

Order placement started
Food preparation started
Delivery started
Order placed successfully
Food delivered successfully
Food prepared successfully


```
23         try {
24             Thread.sleep(600);
25         } catch (InterruptedException e) {
26             Thread.currentThread().interrupt();
27         }
28         System.out.println("Food delivered successfully");
29     });
30     order.start();
31     preparation.start();
32     delivery.start();
33     try {
34         order.join();
35         preparation.join();
36         delivery.join();
37     } catch (InterruptedException e) {
38         Thread.currentThread().interrupt();
39     }
40     System.out.println("All tasks completed.");
41 }
42 }
```

 Run

 Save

Output

Order placement started

Food preparation started

Delivery started

Order placed successfully

Food delivered successfully

Food prepared successfully

Output

Food preparation started

Delivery started

Order placed successfully

Food delivered successfully

Food prepared successfully

All tasks completed.